(i) the probability for Mr.Arivazhagan is appointed as the Physical Teacher can be calculated applying Theorem8.7 as

$$P(A/MD) = \frac{P(MD/A)P(A)}{P(MD/A)P(A)+P(MD/B)P(B)+P(MD/C)P(C)}$$

$$= \frac{0.42 \times 0.45}{(0.42 \times 0.45)+(0.40 \times 0.28)+(0.48 \times 0.27)}$$

$$= \frac{0.189}{0.4306}$$

$$\therefore \quad P(A/MD) = 0.4389.$$

(ii) the probability for Mr.Ilavarasan is appointed as the Physical Teacher can be calculated applying Theorem8.7 as

$$P(B/MD) = \frac{P(MD/B)P(B)}{P(MD/A)P(A)+P(MD/B)P(B)+P(MD/C)P(C)}$$

$$= \frac{0.40 \times 0.28}{(0.42 \times 0.45)+(0.40 \times 0.28)+(0.48 \times 0.27)}$$

$$\therefore \quad P(B/MD) = \frac{0.112}{0.4306}$$

$$= 0.2601.$$

(iii) the probability for Mr.Anbarasan is appointed as the Physical Teacher can be calculated applying Theorem8.7 as

$$P(C|MD) = \frac{P(MD/C)P(C)}{P(MD/A)P(A)+P(MD/B)P(B)+P(MD/C)P(C)}$$

$$= \frac{0.4 \times 0.27}{(0.42 \times 0.45)+(0.40 \times 0.28)+(0.28 \times 0.27)}$$

$$= \frac{0.1296}{0.4306}$$

$$P(C|MD) = 0.3010.$$

Example 8.22

Given three identical boxes I, II and III each containing two coins. In box I, both coins are gold coin, in box II, both are silver coins and in the box III, there is one gold and one silver coin. A person chooses a box at random and takes out a coin. If the coin is of gold, what is the probability that the other coin in the box is also of gold.

Solution:

In fig 8.11 given below if yellow colour denotes gold coin and grey colour denotes silver coin then: